AN EXCEL BASED INDIAN TRAFFIC VIOLATIONS ANALYSIS

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SI NO	TITLE	PAGE NO
1	INTRODUCTION	3
2	DATASET DESCRIPTION	4 -6
3	DATA CLEANING	7 - 8
4	PIVOT TABLE AND CHART	9 -16
5	CONCLUSION	17

1. INTRODUCTION

This project aims to analyze the Indian Traffic Violations Dataset using Microsoft Excel and Business Analytics techniques. The main goal is to understand traffic violation patterns by studying factors such as violation type, location, vehicle type, driver age, weather, road condition, fine amount, payment status, and other enforcement-related variables.

The analysis helps identify which types of violations occur most frequently, where violations are concentrated, how much fine is generated, and which factors are useful for traffic-management decision making. Pivot Tables, Pivot Charts, calculated columns, and dashboard visualizations are used to convert raw traffic records into meaningful business insights.

2. DATASET DESCRIPTION

	A	B	C	D	E	F	G	H	I
1	Violation_ID	Violation_Type	Fine_Amount	Location	Date	Vehicle_Type	Vehicle_Model_Year	Registration_State	Driver_Age
2	VLT100000	Overloading	4544	Karnataka	1/1/2023	Car	2012	West Bengal	25
3	VLT100001	Driving Without License	2776	Punjab	1/2/2023	Scooter	2010	Tamil Nadu	32
4	VLT100002	Using Mobile Phone	4785	Maharashtra	1/3/2023	Scooter	2006	Tamil Nadu	67
5	VLT100003	No Seatbelt	1138	Uttar Pradesh	1/4/2023	Car	1996	Uttar Pradesh	46
6	VLT100004	Over-speeding	1610	Karnataka	1/5/2023	Truck	2016	Delhi	63
7	VLT100005	Over-speeding	4636	Maharashtra	1/6/2023	Bus	2013	Gujarat	33
8	VLT100006	Wrong Parking	2793	West Bengal	1/7/2023	Scooter	2018	Delhi	42
9	VLT100007	Over-speeding	3965	Delhi	1/8/2023	Truck	2009	Punjab	28
10	VLT100008	No Seatbelt	4497	Uttar Pradesh	1/9/2023	Bus	2021	Maharashtra	34
11	VLT100009	Using Mobile Phone	374	West Bengal	1/10/2023	Car	1997	Uttar Pradesh	75
12	VLT100010	Using Mobile Phone	458	Tamil Nadu	1/11/2023	Bus	2005	West Bengal	28
13	VLT100011	Overloading	2236	West Bengal	1/12/2023	Bus	1997	Maharashtra	22
14	VLT100012	No Helmet	915	Maharashtra	1/13/2023	Scooter	2021	Maharashtra	54
15	VLT100013	Signal Jumping	4994	Delhi	1/14/2023	Car	2021	Karnataka	19
16	VLT100014	No Seatbelt	4098	Gujarat	1/15/2023	Truck	2022	Gujarat	47
17	VLT100015	No Seatbelt	1837	West Bengal	1/16/2023	Bus	2011	Gujarat	18
18	VLT100016	No Seatbelt	4646	Punjab	1/17/2023	Auto Rickshaw	2017	Punjab	45
19	VLT100017	No Helmet	4336	Tamil Nadu	1/18/2023	Truck	1996	Uttar Pradesh	56
20	VLT100018	Using Mobile Phone	3715	Punjab	1/19/2023	Auto Rickshaw	1995	Maharashtra	22
21	VLT100019	Signal Jumping	883	Karnataka	1/20/2023	Bus	2014	Gujarat	51
22	VLT100020	Wrong Parking	2727	Gujarat	1/21/2023	Auto Rickshaw	1996	Uttar Pradesh	19
23	VLT100021	Drunk Driving	1502	Punjab	1/22/2023	Scooter	2003	Karnataka	63
24	VLT100022	Signal Jumping	1665	Tamil Nadu	1/23/2023	Bike	2016	West Bengal	64
25	VLT100023	No Helmet	3951	Punjab	1/24/2023	Truck	2007	Delhi	32

Column Name	Description	Data Type
Violation_ID	Unique identifier assigned to each traffic violation.	String
Violation_Type	Type of traffic violation committed.	String
Fine_Amount	Amount of fine imposed for the violation.	Integer
Location	State/location where the violation occurred.	String
Date	Date on which the violation occurred.	Date
Months	Month in which the violation occurred.	String
Vehicle_Type	Type of vehicle involved in the violation.	String

Vehicle_Model_Year	Manufacturing/model year of the vehicle.	Integer
Registration_State	State in which the vehicle is registered.	String
Driver_Age	Age of the driver at the time of violation.	Integer
Driver_Gender	Gender of the driver.	String
License_Type	Type of driving license held by the driver.	String
Penalty_Points	Number of penalty points assigned for the violation.	Integer
Weather_Condition	Weather condition when the violation occurred.	String
Road_Condition	Road condition when the violation occurred.	String
Issuing_Agency	Agency that issued the traffic violation.	String
License_Validity	Validity status of the driver's license.	String
Number_of_Passengers	Number of passengers in the vehicle.	Integer
Helmet_Worn	Indicates whether the driver was wearing a helmet.	String
Seatbelt_Worn	Indicates whether a seatbelt was worn.	String
Traffic_Light_Status	Traffic light status at the time of violation.	String
Speed_Limit	Posted speed limit at the violation location.	Integer

Recorded_Speed	Speed recorded for the vehicle.	Integer
Alcohol_Level	Alcohol concentration measured for the driver.	Decimal/Float
Breathalyzer_Result	Result of the breathalyzer test.	String
Towed	Indicates whether the vehicle was towed.	String
Fine_Paid	Indicates whether the imposed fine was paid.	String
Payment_Method	Method used to pay the fine.	String
Court_Appearance_Required	Indicates whether court appearance is required.	String
Previous_Violations	Number of previous traffic violations by the driver.	Integer
Year	Year in which the violation occurred.	Integer
Age Category	Age group/category assigned to the driver.	String

Dataset Source : Kaggle

Total Rows:4000

Total Columns:32

3. DATA CLEANING

1. Importing Data into Excel:-

The dataset was first imported into Microsoft Excel.The Excel workbook containing the Indian Traffic Violations dataset was opened,and the data was placed in a worksheet for further cleaning and analysis.

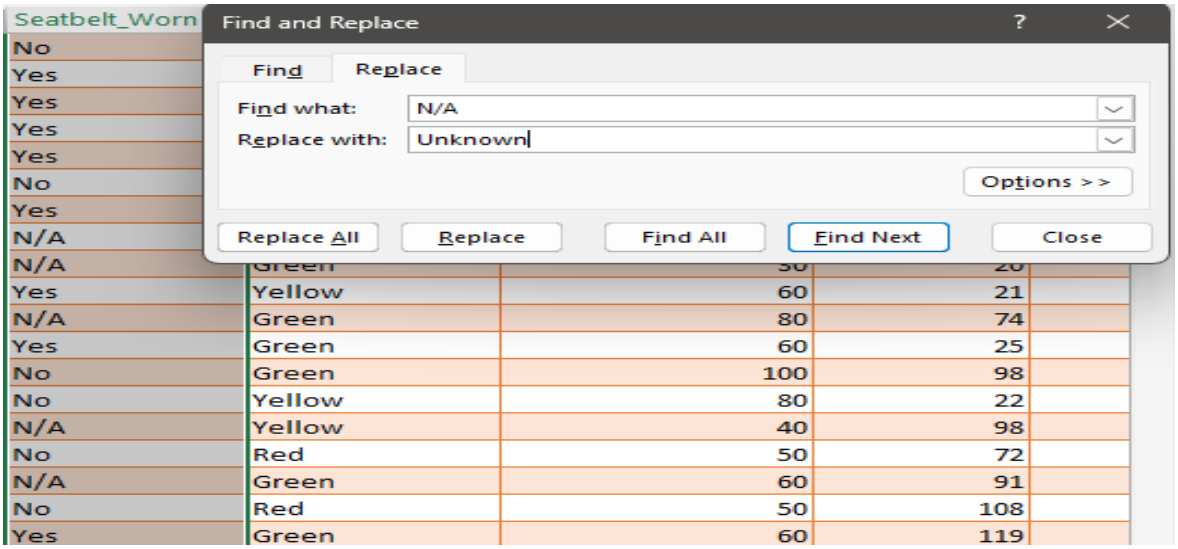
2. Alt +H + O + I:- Autofit Column Width.

3. Alt + H + O +A:- Autofit Row Height.

4. Checking Missing Values:-The dataset was checked for blank or missing values. Filters were used to identify empty cells in different columns.Helmet Worn and Seatbelt Worn contained missing values.

5. Handling Missing Values:-

- **Seatbelt Worn:-** Fill with Unknown.
- **Helmet Worn:-** Fill with Unknown.



Ctrl + H → Find what: `N/A` → Replace with: `Unknown` → Replace All

The **N/A values** in the Seatbelt_Worn and Helmet_Worn columns were replaced with **"Unknown"** because the available data does not specify whether the seatbelt or helmet was worn.

6. Remove Unwanted Columns:-

- **Vehicle Color -** Not required for the main traffic violation analysis.
- **Comments** - Free-text information and not required for analysis.
- **Officer ID** - Officer-level identification is not required for this project.
- **Time** - Time-level analysis is outside the scope of the project.

7. Extract Year , Month:-

The **Year** and **Month** are extracted from the **Date** column to make it easier to analyze traffic violations by year and month.

- **Year:** Extract the year from the Date column.
- **Month:** Extract the month from the Date column.

Year: =YEAR([@Date])
Month: =TEXT([@Date],"mmmm")

8. Age Category:-

=IFS([@[Driver_Age]]<18,"Minor",[@[Driver_Age]]<=25,"Young Adult",[@[Driver_Age]]<=40,"Adult",[@[Driver_Age]]<=60,"Middle Aged",[@[Driver_Age]]>60,"Senior")

PIVOT TABLE &CHART

1. What is the total number of violations by violation type?

Row Labels	Count of Violation_ID
Driving Withou	451
Drunk Driving	488
No Helmet	463
Over-speeding	448
Wrong Parking	454
Grand Total	**2304**



INTERPRETATION:

Drunk Driving has the highest number of violations at **488**, followed by No Helmet (**463**) and Wrong Parking (**454**). Over-speeding has the lowest number at **448**. This indicates that drunk driving is the most frequently recorded violation among the categories shown.

2. Which violation type generates the highest total fine amount?

Row Labels	Sum of Fine_Amount
Driving Withou	1165765
Drunk Driving	1277302
No Helmet	1159607
No Seatbelt	1130823
Signal Jumping	1116155
Grand Total	**5849652**

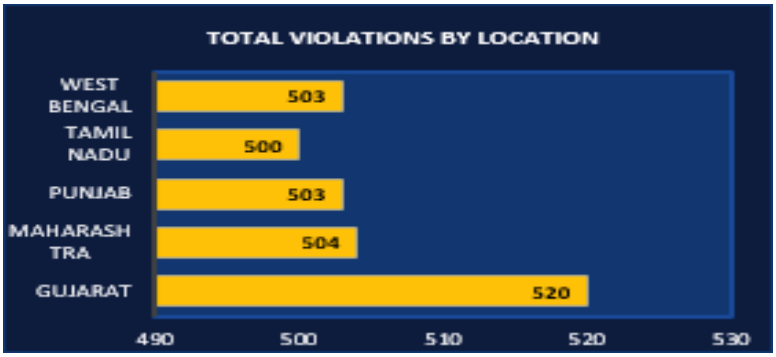


INTERPRETATION:

Drunk Driving generates the highest total fine amount of approximately **₹12.77 lakh**. Signal Jumping generates the lowest total fine amount at approximately **₹11.16 lakh**. This suggests that drunk-driving violations have a significant financial impact.

3. Which location has the highest number of traffic violations?

Row Labels	Count of Violation_ID
Gujarat	520
Maharashtra	504
Punjab	503
Tamil Nadu	500
West Bengal	503
Grand Total	**2530**



INTERPRETATION:

Gujarat has the highest number of violations with **520**, followed by Maharashtra with **504**. Tamil Nadu has the lowest among the locations shown, with **500** violations. The difference is relatively small, indicating that violations are fairly evenly distributed across these locations.

4. How do traffic violations vary by month?

Row Labels	Count of Violation_ID
January	341
February	311
March	341
April	330
May	341
June	330
July	341
August	341
September	330
October	341
November	330
December	323
Grand Total	**4000**

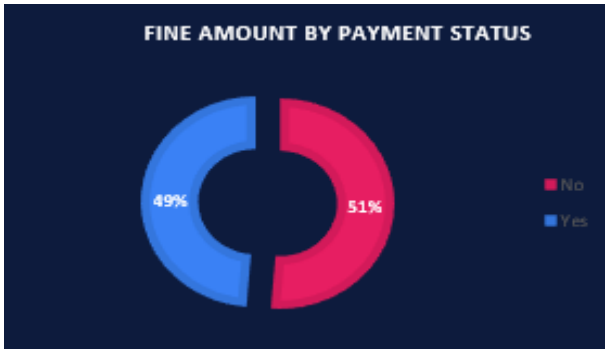


INTERPRETATION:

The monthly trend shows that violations remain relatively stable throughout the year, mostly ranging between **330 and 341**. **January, March, June, July, August, September and October** show around 341 violations, while **December has the lowest value at 323**. Overall, there is no major seasonal fluctuation.

5. What percentage of fines are paid?

Row Labels	Sum of Fine_Amount
No	5187461
Yes	4931824
Grand Total	**10119285**

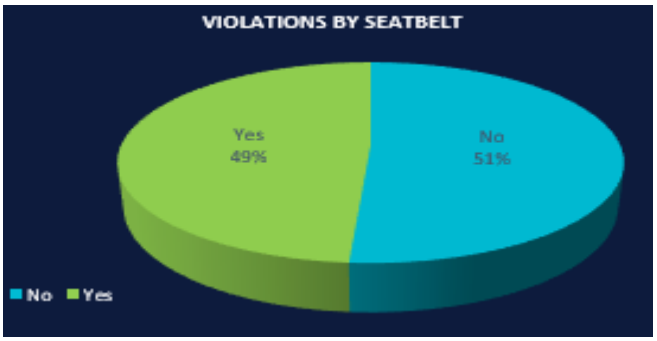


INTERPRETATION:

The chart shows that **49% of fines are paid**, while **51% remain unpaid**. This indicates that unpaid fines slightly exceed paid fines, suggesting that improving fine collection and payment compliance could be an important area for enforcement.

6. What is the seatbelt compliance rate?

Row Labels	Count of Violation_ID
No	1385
Yes	1323
Grand Total	**2708**



INTERPRETATION:

Approximately **49% of drivers were wearing seatbelts**, while **51% were not wearing them**. The results indicate that seatbelt non-compliance is slightly higher than compliance, highlighting the need for greater awareness and enforcement of seatbelt regulations.

7. Which registration state has the highest number of violations?

Row Labels	Count of Violation_ID
Delhi	498
Maharashtra	524
Punjab	523
Uttar Pradesh	501
West Bengal	504
Grand Total	**2550**

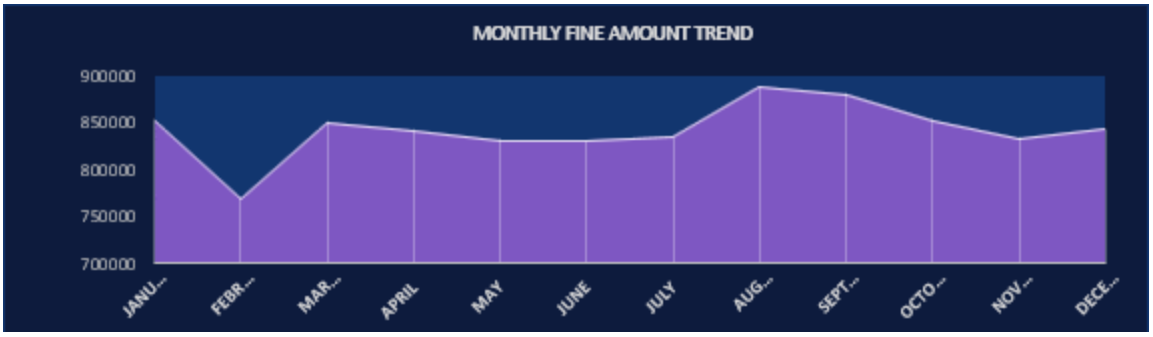


INTERPRETATION:

Maharashtra has the highest number of violations with **524**, closely followed by **Punjab with 523**. Delhi has the lowest number among the states shown, with **498**.

8. Which month has the highest total fine amount?

Row Labels	Sum of Fine_Amount
January	854058
February	770184
March	851022
April	843313
May	832699
June	831166
July	835647
August	888251
September	880834
October	854011
November	834241
December	843859
Grand Total	**10119285**

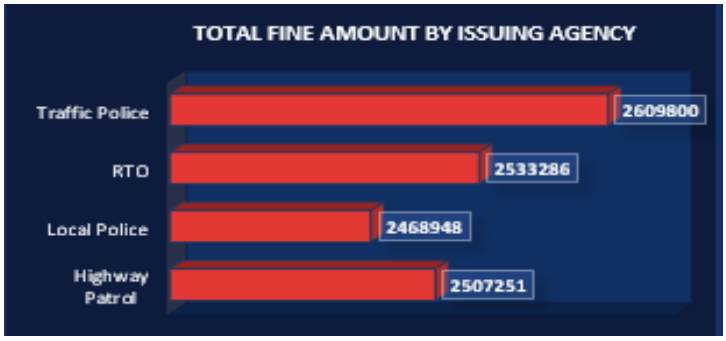


INTERPRETATION:

August records the highest monthly fine amount at approximately **₹8.9 lakh**. February has the lowest, at approximately **₹7.7 lakh**. This indicates that fine collection is highest during August.

9. Which issuing agency collected the highest total fine amount?

Row Labels	Sum of Fine_Amount
Highway Patro	2507251
Local Police	2468948
RTO	2533286
Traffic Police	2609800
Grand Total	**10119285**

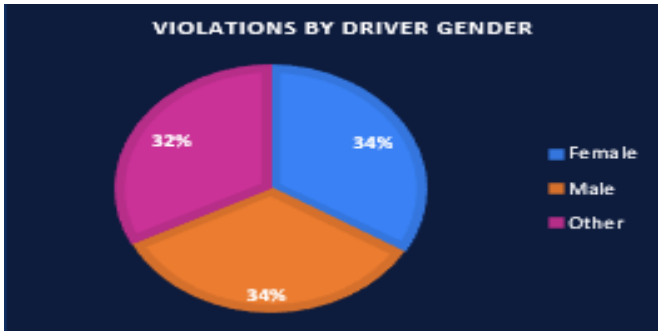


INTERPRETATION:

The **Traffic Police** collected the highest total fine amount at approximately **₹26.1 lakh**. The **Local Police** collected the lowest, at approximately **₹24.7 lakh**. This suggests that Traffic Police handle a large share of traffic-violation enforcement.

10. What is the distribution of violations by driver gender?

Row Labels	Count of Violation_ID
Female	1343
Male	1359
Other	1298
Grand Total	**4000**

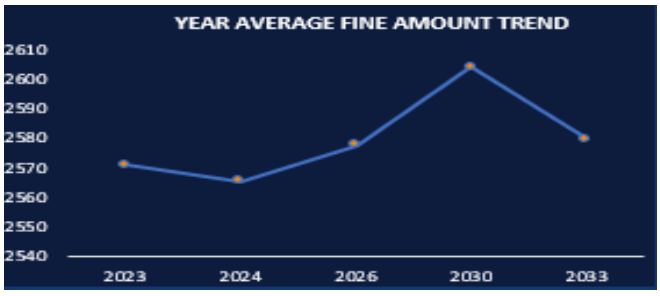


INTERPRETATION:

Female and Male drivers each account for approximately 34% of violations, while **Other** accounts for approximately **32%**. The distribution is relatively balanced across the three categories.

11. How does the average fine amount change by year?

Row Labels	Average of Fine_Amount
2023	2571
2024	2565
2026	2578
2030	2604
2033	2580
Grand Total	**2579**

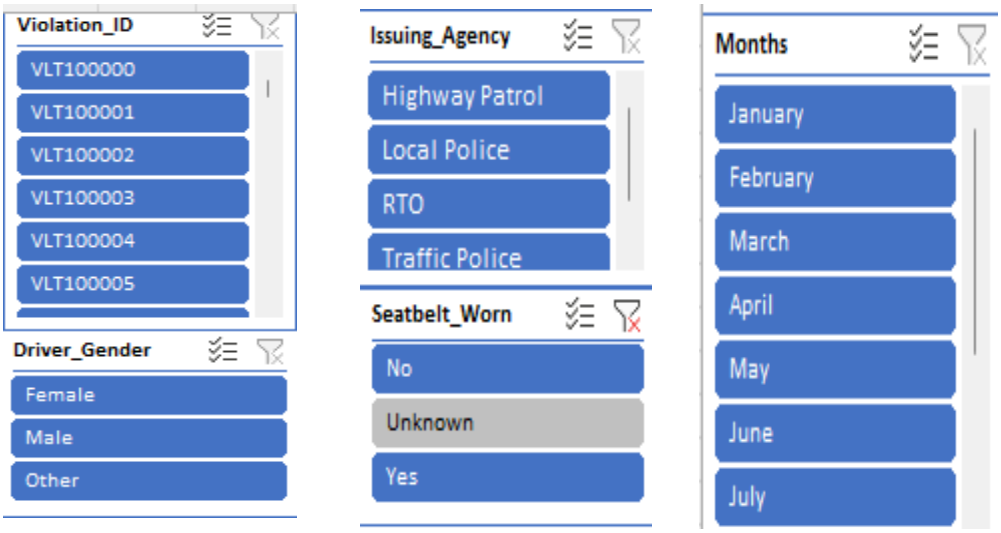


INTERPRETATION:-

The average fine amount decreases slightly from **2023 to 2024**, then increases through **2030**, where it reaches its highest level of approximately **₹2,605**. It then decreases again in **2033** to approximately **₹2,580**. Overall, the average fine shows some fluctuation rather than a consistent upward or downward trend.

SLICER

A **Slicer** is an interactive filtering tool in Excel that allows users to **quickly filter Pivot Tables and Pivot Charts** by selecting specific categories.



CONCLUSION

The Indian Traffic Violations analysis provides useful insights into traffic offences, fines, driver behaviour, and locations. **Drunk Driving** is a major violation, while **Maharashtra and Punjab** record comparatively higher violations.

The analysis also shows that **August has a high fine amount**, while **51% of fines are unpaid** and **51% of drivers are not wearing seatbelts**. Overall, the project demonstrates how Excel tools can help identify traffic patterns and support better **road-safety and enforcement decisions**.